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Project: Bank Customer Churn Prediction

Bank Customer Churn Prediction

Problem Statement

Customer churn is a major challenge for banks because losing customers directly impacts revenue and business growth. The objective of this project is to build a Machine Learning model that predicts whether a customer is likely to leave the bank based on demographic and account-related information.

Dataset Description

Dataset Name: Churn_Modelling.csv

Total Records: 10,000

Features:

- CreditScore
- Geography
- Gender
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary

Target Variable:

- Exited
 - 1 = Customer Left Bank
 - 0 = Customer Stayed

Removed Columns:

- RowNumber
- CustomerId
- Surname

Methodology

1. Data Cleaning

- Checked missing values
- Removed duplicate records
- Removed unnecessary columns
- Performed consistency checks

2. Feature Engineering

Created new features:

1. BalanceSalaryRatio
2. ProductsPerTenure
3. AgeGroup

3. Data Preprocessing

- Train-Test Split (80:20)
- Encoding

-
- Scaling

4. Model Training

Three Machine Learning Models were trained:

1. Logistic Regression
2. Decision Tree
3. Random Forest

5. Model Evaluation

Metrics Used:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

6. Deployment

- Saved model using Joblib
- Built Streamlit Web Application

Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8360 (83.60%)	0.6816	0.3104	0.4266	0.6375
Decision Tree	0.7895 (78.95%)	0.4671	0.5064	0.4860	0.6826
Random Forest	0.8610 (86.10%)	0.7386	0.4529	0.5615	0.7069

Best Model:

Random Forest Classifier

Reason:

Highest accuracy and better overall performance.

Screenshots

Prediction Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
TERMINAL
& C:\Users\YASMEEN\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/YASMEEN/OneDrive/Desktop/ML intern/Assignment/Bank_Cus...
tomer_Churn/src/train_model.py"

Dataset Shape
(10000, 14)

Data Types
RowNumber      int64
CustomerId     int64
Surname        object
CreditScore    int64
Geography      object
Gender         object
Age            int64
Tenure         int64
Balance        float64
NumOfProducts int64
HasCrCard      int64
IsActiveMember int64
EstimatedSalary float64
Exited         int64
dtype: object

Missing Values
RowNumber      0
CustomerId     0
Surname        0
CreditScore    0
Geography      0
Gender         0
Age            0
Tenure         0
Balance        0
NumOfProducts  0
HasCrCard      0
IsActiveMember 0
EstimatedSalary 0
Exited         0
dtype: int64

Duplicate Rows
0

Statistical Summary
      RowNumber  CustomerId  CreditScore  Age  Tenure  ...  NumOfProducts  HasCrCard  IsActiveMember  EstimatedSalary  Exited
count  10000.00000  1.000000e+04  10000.00000  10000.00000  10000.00000  ...  10000.00000  10000.00000  10000.00000  10000.00000  10000.00000
mean    5000.50000  1.569094e+07  650.528800  38.921800  5.012800  ...  1.530200  0.70550  0.515100  57510.492818  0.203700
std    2886.89568  7.193619e+04  96.653299  10.487806  2.892174  ...  0.581654  0.45584  0.499797  57510.492818  0.402769
min       1.00000  1.556570e+07  350.000000  18.000000  0.000000  ...  1.000000  0.00000  0.000000  11.580000  0.000000
```

```
TERMINAL
tomer_Churn/src/train_model.py"

mean    5000.50000  1.569094e+07  650.528800  38.921800  5.012800  ...  1.530200  0.70550  0.515100  57510.492818  0.203700
std    2886.89568  7.193619e+04  96.653299  10.487806  2.892174  ...  0.581654  0.45584  0.499797  57510.492818  0.402769
min       1.00000  1.556570e+07  350.000000  18.000000  0.000000  ...  1.000000  0.00000  0.000000  11.580000  0.000000
25%    2500.75000  1.562853e+07  584.000000  32.000000  3.000000  ...  1.000000  0.00000  0.000000  51002.110000  0.000000
50%    5000.50000  1.569074e+07  652.000000  37.000000  5.000000  ...  1.000000  1.00000  1.000000  100193.915000  0.000000
75%    7500.25000  1.575323e+07  718.000000  44.000000  7.000000  ...  2.000000  1.00000  1.000000  149388.247500  0.000000
max   10000.00000  1.581569e+07  850.000000  92.000000  10.000000  ...  4.000000  1.00000  1.000000  199992.480000  1.000000

[8 rows x 11 columns]

-----
Logistic Regression
-----
Accuracy: 0.836
Precision: 0.6815642458100558
Recall: 0.3104325699745547
F1 Score: 0.42657342657342656
ROC AUC: 0.63748137521752

Confusion Matrix
[[1550  57]
 [ 271 122]]

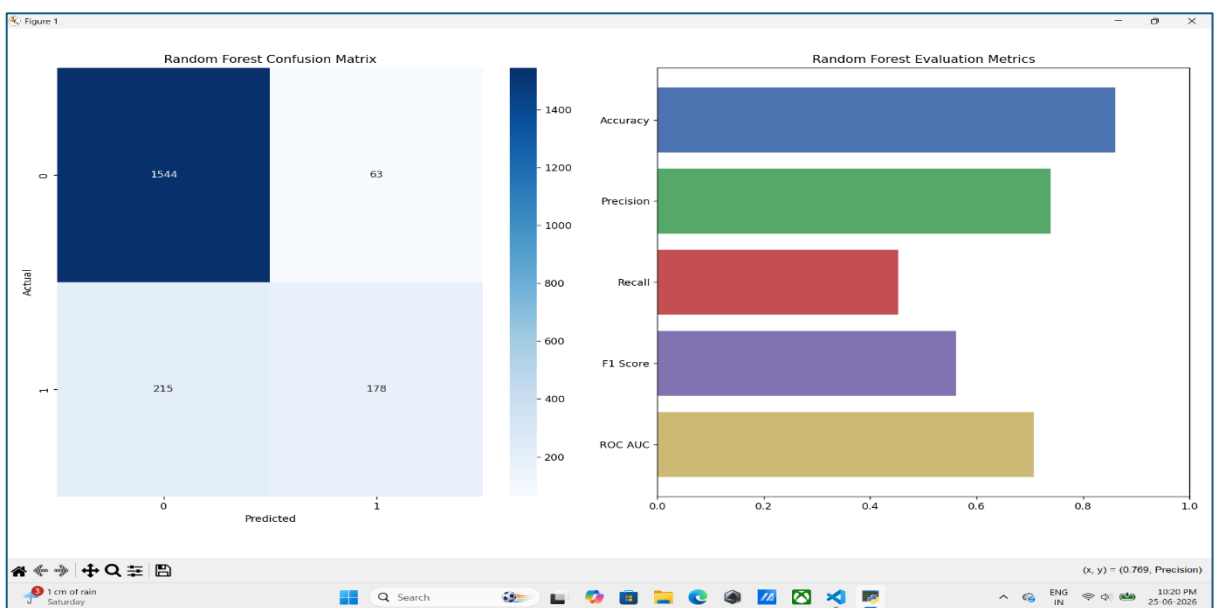
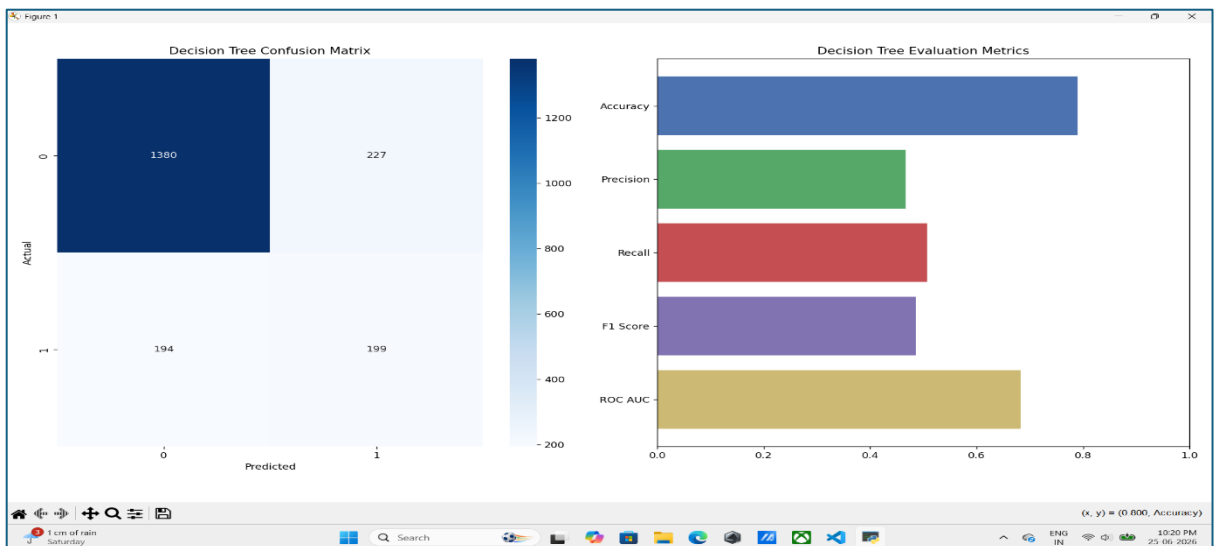
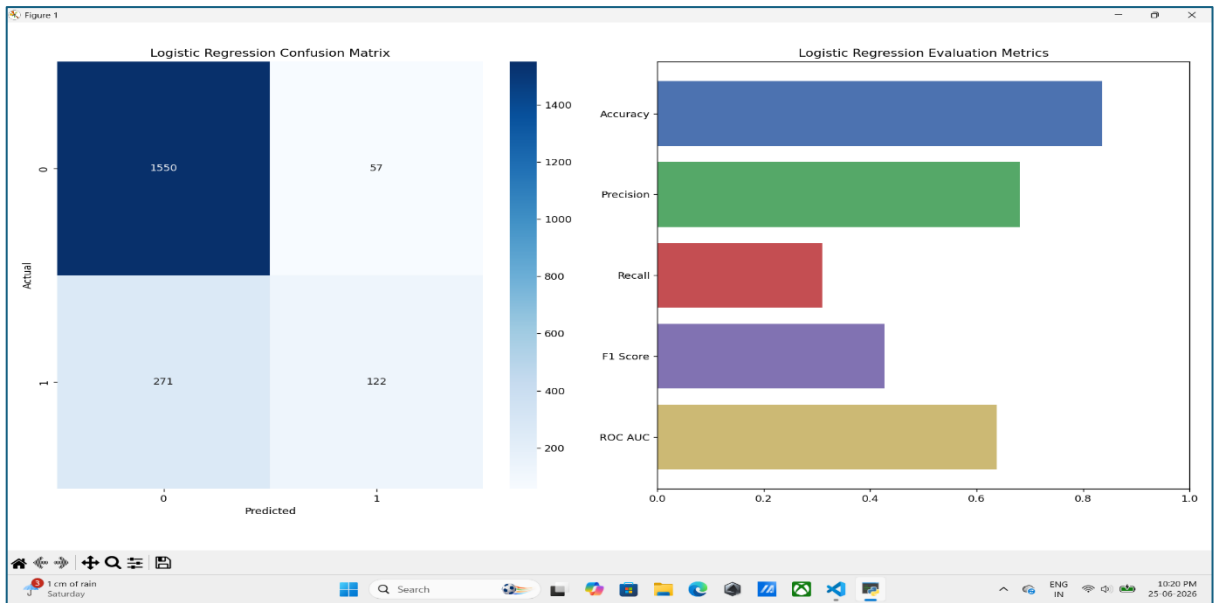
-----
Decision Tree
-----
Accuracy: 0.7895
Precision: 0.4669811320754717
Recall: 0.5038167938931297
F1 Score: 0.4847001223990208
ROC AUC: 0.6815910354033166

Confusion Matrix
[[1381 226]
 [ 195 198]]

-----
Random Forest
-----
Accuracy: 0.861
Precision: 0.7395892116182573
Recall: 0.4529262086513995
F1 Score: 0.5615141955835962
ROC AUC: 0.7068613619485995

Confusion Matrix
[[1544  63]
 [ 215 178]]

Saved Successfully
PS C:\Users\YASMEEN\OneDrive\Desktop\ML intern>
```



Streamlit Home Page:



Bank Customer Churn Prediction

Enter customer details

Credit Score

600

- +

Geography

Germany

▼

Gender

Male

▼

Age

35

- +

Tenure

5

- +

Balance

0.00

- +

Number Of Products

1

- +

Has Credit Card

0

▼

Is Active Member

0

▼

Estimated Salary

0.00

- +

Predict

Customer Will Stay



Bank Customer Churn Prediction

Enter customer details

Credit Score

600

-

+

Geography

Germany

▼

Gender

Male

▼

Age

70

-

+

Tenure

5

-

+

Balance

0.00

-

+

Number Of Products

1

-

+

Has Credit Card

1

▼

Is Active Member

0

▼

Estimated Salary

0.00

-

+

Predict

Customer Will Leave Bank

How to Run

Train Model

```
python train_model.py
```

Run Streamlit Application

```
streamlit run app.py
```

Open Browser

<http://localhost:8501>

Project Structure

Bank_Churn_Project/

├── Churn_Modelling.csv

├── train_model.py

├── app.py

├── model.pkl

├── preprocessor.pkl

├── requirements.txt

└── README.md

Technology Stack:

- Python
- Pandas
- NumPy
- Scikit-Learn
- Streamlit
- Joblib

Conclusion

This project successfully predicts whether a customer is likely to leave the bank or stay. Different machine learning models were trained and evaluated, and Random Forest provided the best results. The model was deployed using Streamlit, allowing users to make predictions through a simple web interface. The system can help banks improve customer retention and make better business decisions.